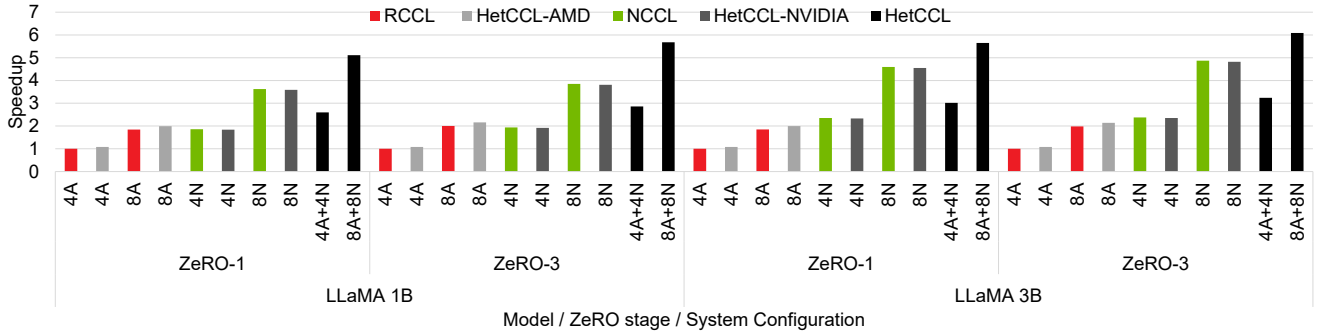
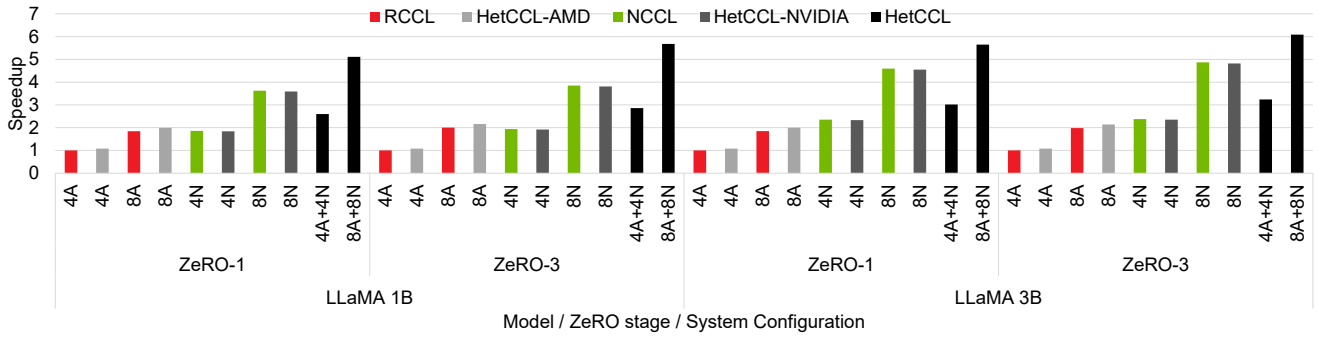
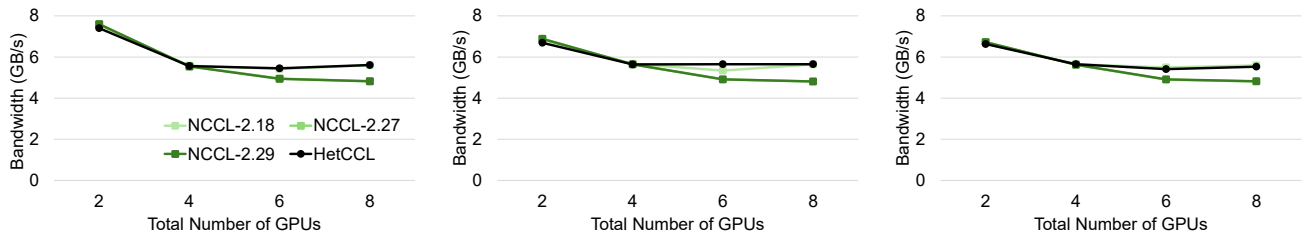


(a) GPT training



(b) LLaMA training

Figure 1: Speedup in LLM training. 4A/8A and 4N/8N represent training runs on 4/8 AMD GPUs and 4/8 NVIDIA GPUs, respectively. 4A+4N/8A+8N denotes a heterogeneous setting with 4/8 AMD and 4/8 NVIDIA GPUs.



(a) All-Reduce (b) All-Gather (c) Reduce-Scatter

Figure 2: Comparison of collective communication performance across NCCL versions 2.18, 2.27, and 2.29.

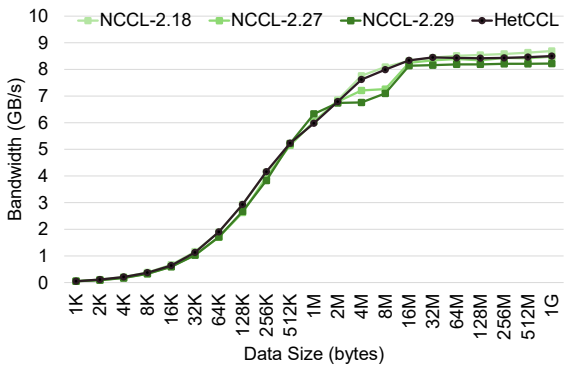


Figure 3: Comparison of point-to-point communication performance across NCCL versions 2.18, 2.27, and 2.29.